Project proposal (Synopsis) of

Bachelor of Computer Application

On

PetZio

Everything your pet needs, thoughtfully organized in one place.

To

Project Co-Ordinator (BCA)

INDIRA GANDHI NATIONAL OPEN UNIVERSITY

SUBMITTED BY

ENROLMENT NO. : 2351641695

NAME : TANNU

UNDER GUIDANCE OF : AKHILESH MISHRA

Table of Contents

1. Title of Project

PetZio

(An Online Pet Product Store)

2. Introduction

PetZio is a web-based application that focuses on simplifying the process of purchasing pet supplies through an online platform. The system aims to bring together a wide range of pet-related products in one place, making it easier for pet owners to access essential items for their pets. The platform is designed to support a structured and user-friendly digital shopping experience.

The application is intended to serve different types of users, including customers, vendors, and administrators, each with clearly defined roles. Customers are able to browse products, view detailed information, and place orders based on their requirements. Vendors manage product listings and availability, while administrators oversee system operations, user management, and overall platform control. This role-based structure helps maintain smooth coordination and efficient management of the system.

PetZio is designed with scalability and flexibility in mind to accommodate future enhancements. The system supports secure transactions, organized data handling, and an intuitive interface to meet the growing needs of the pet supplies domain.

PetZio also emphasizes efficient data organization and smooth interaction between different system components. The platform ensures reliable information flow between users, products, and orders. This structured approach helps maintain consistency, accuracy, and ease of future expansion.

3. Objectives

1) **To provide a centralized online platform for pet supplies**

   The system aims to offer a single platform where a wide variety of pet-related products can be displayed and accessed. It focuses on reducing the effort required to search for pet supplies from multiple sources. The platform is intended to improve convenience for pet owners. This objective supports a simplified and organized shopping experience.

2) **To support role-based access for different users**

   The application is designed to handle multiple user roles such as customers, vendors, and administrators. Each role is assigned specific permissions and responsibilities within the system. This separation helps in maintaining proper control and accountability. It also ensures smooth and secure operation of the platform.

3) **To enable efficient product and order management**

   The system focuses on structured handling of product listings, pricing, and availability. It allows proper tracking of customer orders and related details. This objective helps in maintaining accuracy and reducing manual errors. Efficient management improves overall system reliability.

4) **To ensure secure handling of user and transaction data**

   The application aims to protect user information through secure data handling mechanisms. It focuses on maintaining confidentiality and integrity of customer and vendor data. Secure authentication and authorization processes are emphasized. This objective helps build trust among users of the system.

5) **To design a user-friendly and responsive interface**

The platform focuses on providing an easy-to-navigate and visually clear interface. It aims to enhance usability across different devices and screen sizes. Simple layouts and smooth interaction are prioritized. This objective improves overall user satisfaction and accessibility.

6) **To support scalability and future enhancement of the system**

The system is structured to allow easy addition of new features in the future. It focuses on flexible design and modular architecture. This objective ensures that the application can adapt to growing user needs. It also supports long-term maintenance and expansion of the platform.

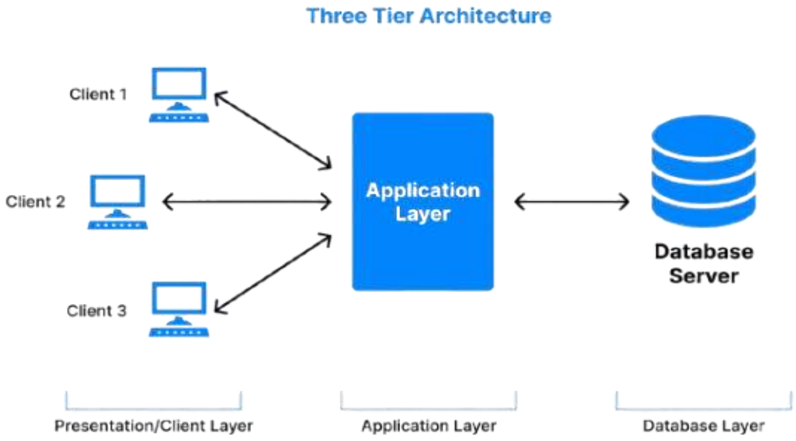
4. Project Category

- The application is Web based using **three-tier architecture**, which divides the system into presentation, application, and data layers. The presentation layer manages user interaction through web interfaces used by customers, vendors, and administrators. The application layer handles business logic such as user authentication, product management, and order processing. The data layer stores and manages information related to users, products, and transactions. This architectural approach helps improve system security, scalability, and clarity in design.

- PetZio is categorized as a web-based e-commerce application that operates using a client–server model. The client component allows users to access the system through a web browser to perform actions such as viewing products and managing accounts. The server component processes these requests and applies the required business rules. Communication between the client and server ensures accurate data exchange and consistent system behaviour. This structure supports centralized control and efficient request handling.

- The system uses a combination of modern technologies to maintain smooth operation and flexibility. User interfaces are structured using HTML, JavaScript, and Tailwind CSS for responsiveness and ease of use. Server-side processing is handled using Node.js and Express.js to manage application flow and services. Data storage and retrieval are managed through MongoDB with Mongoose for structured data handling.

This setup supports maintainability, performance, and future functional expansion.



➢Reasons For Using Node.js over React: -

- **Broad Industry Acceptance**

Node.js will be commonly used in modern web application development. Its wide adoption will provide access to extensive documentation and community support. This will help during the development of the PetZio platform.

- **Single Programming Language**

The use of JavaScript across the application will simplify development activities. It will reduce complexity by avoiding multiple programming languages. This will make maintenance and future updates easier.

- **High Execution Efficiency**

Node.js will be capable of handling multiple operations efficiently. It will support fast processing of user actions such as product browsing and order placement. This will help maintain responsive application behaviour.

- **Asynchronous Operation Handling**

The system will support asynchronous execution for tasks like data access and background processing. These operations will run without blocking other activities. This will ensure smooth functioning under multiple user requests.

- **MongoDB Capabilities**

Node.js will integrate smoothly with MongoDB using Mongoose for data handling. This will support efficient storage and retrieval of product, user, and order information. It will help maintain flexibility in managing application data.

➢Reasons for using DBMS: -

1. **Flexible Data Structure**
   The application will store diverse data such as user profiles, pet products, orders, and vendor details. A DBMS will allow handling of semi-structured and unstructured data without enforcing rigid table formats. This flexibility will support changes in data requirements as the system evolves.

2. **Ease of Scalability**
   The platform will be expected to handle an increasing number of users and product records over time. A DBMS will support scalable data storage without complex

restructuring. This will help the system grow smoothly with rising data volume.

3. **Efficient Handling of Large Data**

   The system will manage large datasets related to products, transactions, and user activities. A DBMS will process such data efficiently without heavy dependency on joins. This will improve data access speed and overall performance.

4. **Schema Modification Support**

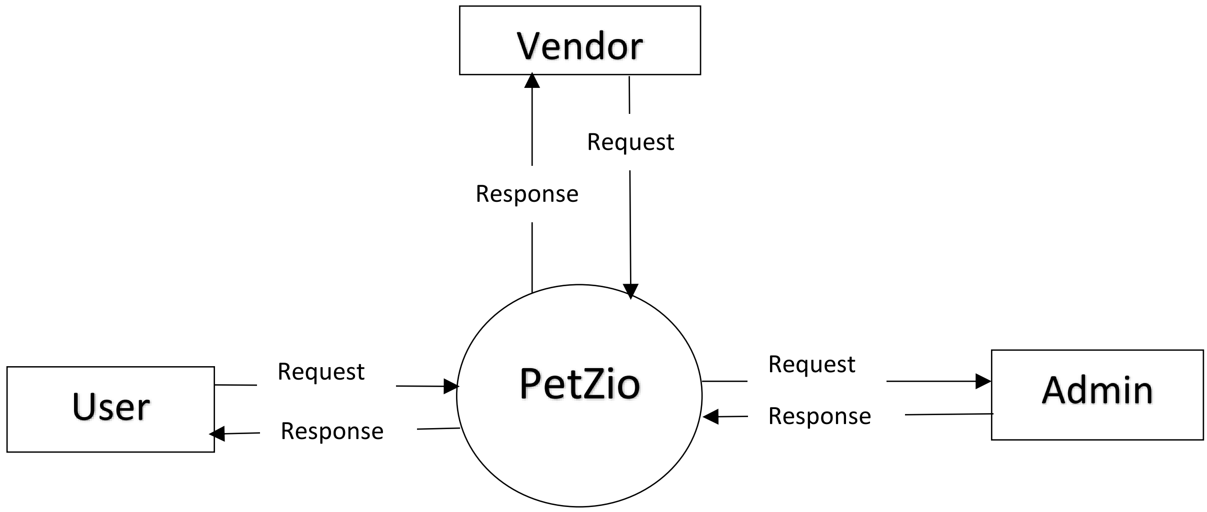
   Application requirements may change during future development and enhancement phases. A DBMS will allow easy modification of data structures without affecting existing data heavily. This will reduce maintenance effort and support system adaptability.

5. **Better Integration with Modern Applications**

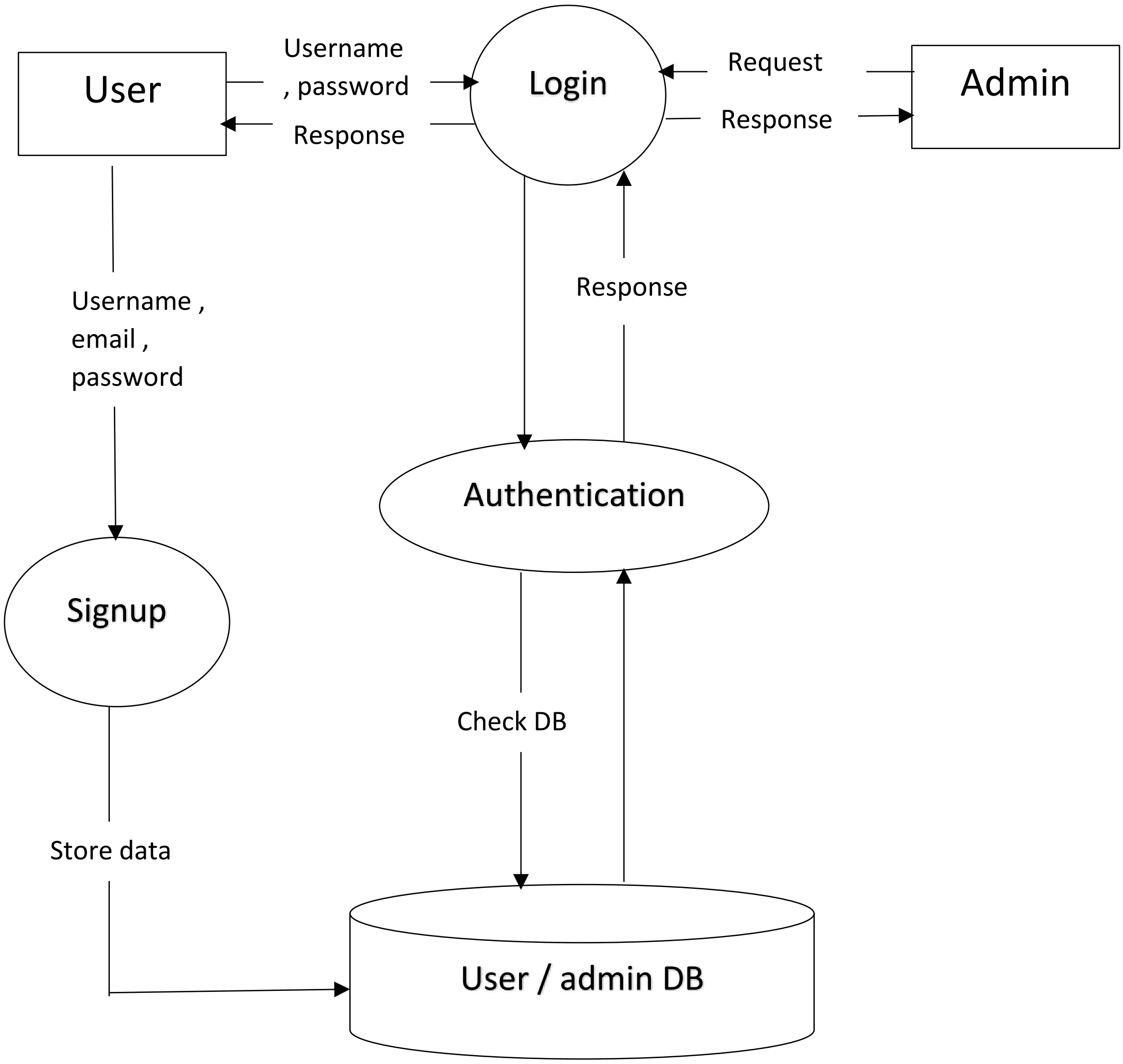
   The application will use modern web technologies and backend frameworks. A DBMS will integrate smoothly with these technologies through flexible data models. This will simplify development and improve data management efficiency.

5. Data Flow Diagram

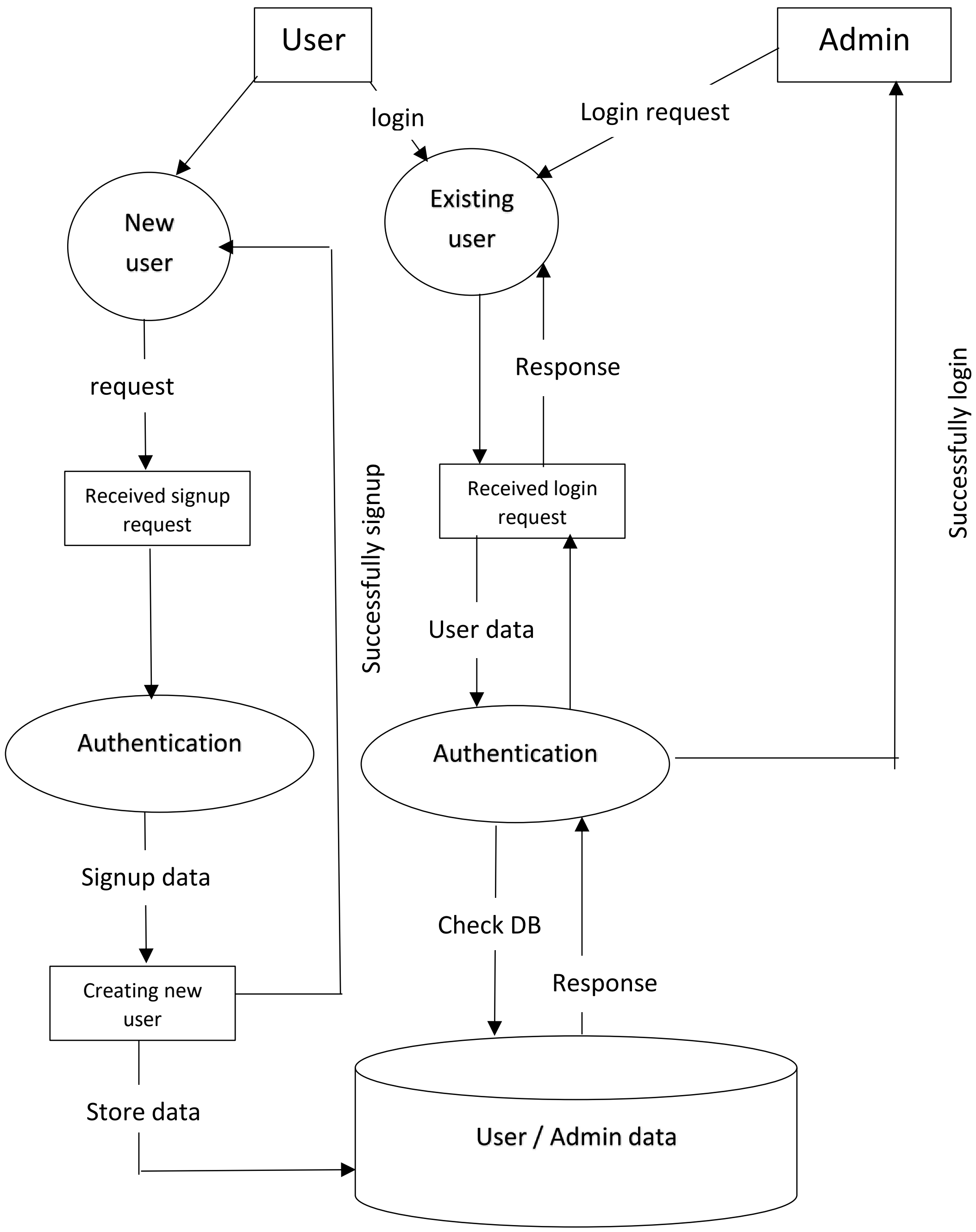
(Level 0)



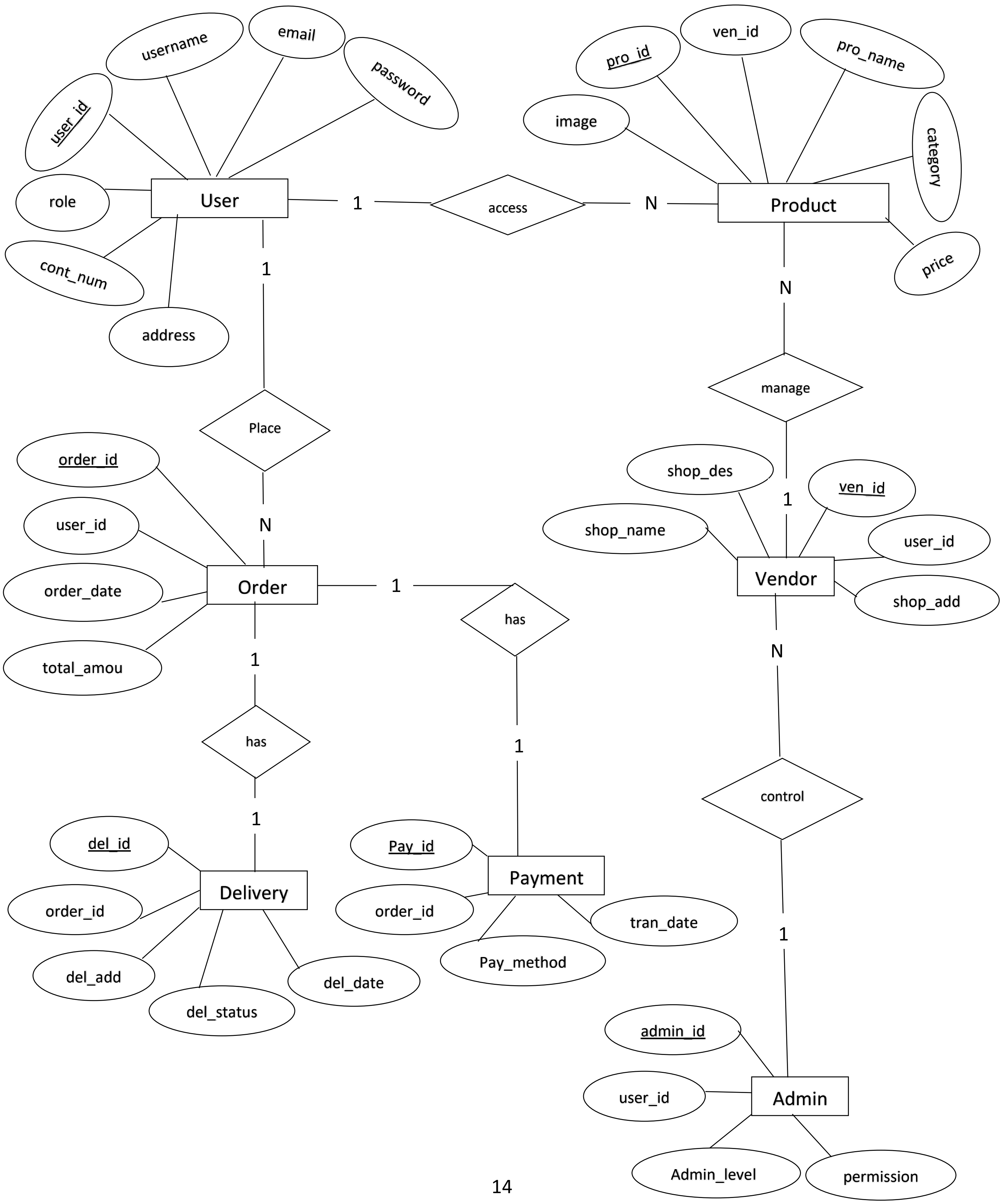
(Level 1)


User
Username
, password
Response
Login
Request
Response
Admin
Username ,
email ,
password
Response
Authentication
Signup
Check DB
Store data
User / admin DB

(Level 2)


User
Admin
login
Login request
New user
Existing user
request
Response
Successfully login
Received signup request
Received login request
Successfully signup
User data
Authentication
Authentication
Signup data
Check DB
Response
Creating new user
Store data
User / Admin data

6. Entity Relation Diagram



7. Data Structure

Table Name : User

Column name	Data Type	Constraints	Description
user_id	ObjectId	primary key	Unique identifier for each user
username	String	required	Login username of the user
email	String	required , unique	Email address of the user
password	String	required	Hashed password for authentication
role	String	required	Defines role (user/admin/vendor)
cont_num	String	required	User contact number
address	String	required	User address number

Table Name : Admin

Column name	Data Type	Constraints	Description
admin_id	ObjectId	primary key	Unique identifier for admin
user_id	ObjectId	Foreign key (user)	Reference to user acting as admin
admin_level	String	required	Level of administrative authority
permission	String	required	List of admin permissions

Table Name : Vendor

Column name	Data Type	Constraints	Description
ven_id	ObjectId	primary key	Unique identifier for vendor
user_id	ObjectId	Foreign key (user)	Reference to user acting as vendor
shop_name	String	required	Vendor shop name
shop_des	String	required	Vendor shop description
shop_add	String	required	Vendor shop address

Table Name : Product

Column name	Data Type	Constraints	Description
pro_id	ObjectId	primary key	Unique identifier for product
ven_id	ObjectId	Foreign key (vendor)	Vendor managing the product
pro_name	String	required	Name of the pet product
category	String	required	Product category
price	Number	required	Product price
image	String	optional	Image URL of product

Table Name : Order

Column name	Data Type	Constraints	Description
order_id	ObjectId	primary key	Unique identifier for order
user_id	ObjectId	Foreign key (user)	User who place the order
order_date	Date	required	Date of order place
total_amou	Number	required	Total order amount

Table Name : Delivery

Column name	Data Type	Constraints	Description
del_id	ObjectId	primary key	Unique identifier for delivery
order_id	ObjectId	Foreign key (order)	Order linked to delivery
del_date	Date	Optional	Date of delivery
del_add	String	required	Address of delivery
del_status	String	required	Current delivery status

Table Name : Payment

Column name	Data Type	Constraints	Description
pay_id	ObjectId	primary key	Unique identifier for payment
order_id	ObjectId	Foreign key (order)	Order linked to payment
tran_date	Date	required	Date of payment
pay_method	String	required	Payment method use

8. Tools / Platform , Hardware and Software

❖ Tools and Platform: -

- **Frontend** : Nextjs , tailwindCSS , GSAP ,ShadCN
- **Backened** : Node.js , express.js
- **Database** : MongoDB with Mongoose
- **IDE** : Virtual Studio Code
- **Deployment** : Vercel for Frontened , Render for Backened , MongoDB Atlas for DB

❖ Hardware Requirement :-

- 8 GB RAM
- Core – i5 CPU
- 128 GB Storage
- Working Internet Connection

❖ Software Requirement :-

- Node.js
- MongoDB
- Git
- NPM
- Postman

9. Number of Modules and Description

1)User Management Module

2)Authentication Module

3)Product Management Module

4)Vendor Management Module

5)Order Management Module

6)Payment Management Module

7)Delivery Management Module

8)Search and Browsing Module

9)Admin Control Module

10) Notification Module

9.1 Login Management Process :-

The **login management process** will control user access to the **PetZio** platform by verifying user credentials. **Users** will enter their username and password to initiate the login process. The system will validate these credentials against stored user data. Only authenticated users will be allowed to proceed further into the application.

After **successful authentication**, role-based access will be applied within the system. Users will be directed to specific dashboards based on their assigned roles such as customer, vendor, or administrator. Unauthorized login attempts will be restricted to protect system resources. This process will help ensure secure and controlled platform usage.

❖Module Description :-

- **User Management Module**

This module will handle user registration and profile maintenance. It will manage personal details of customers, vendors, and administrators. User information updates will be supported when required. The module will ensure organized handling of user data.

- **Authentication Module**

This module will manage login and logout activities within the system. It will verify user credentials during the authentication process. Access rights will be assigned based on user roles. Secure session handling will be maintained through this module.

- **Product Management Module**

This module will manage pet product information displayed on the platform. It will support adding, updating, and viewing product details. Product categories and availability will be handled here. The module will help maintain a structured product catalog.

- **Vendor Management Module**

This module will manage vendor-related information and activities. Vendors will be able to control their product listings through this module. Vendor details will be stored and maintained systematically. It will ensure proper coordination between vendors and the system.

- **Order Management Module**

This module will manage order placement and order records. It will store order details generated by customers. Order status tracking will be supported within the module. It will ensure smooth handling of purchase activities.

- **Payment Management Module**

This module will handle payment-related details for orders. It will manage payment methods and payment status information. Secure handling of transaction records will be ensured. The module will support accurate payment tracking.

- **Delivery Management Module**

This module will manage delivery information for customer orders. It will store delivery address and delivery status details. Order shipment tracking will be supported here. It will assist in timely order fulfilment.

- **Search and Browsing Module**

This module will allow users to search and browse pet products easily. It will support filtering based on categories and preferences. Product viewing features will be managed through this module. It will enhance user navigation experience.

- **Admin Control Module**

This module will provide administrative control over the entire system. Admin users will manage users, vendors, and products through this module. System monitoring activities will be performed here. It will ensure effective system supervision.

- **Notification Module**

This module will manage system-generated notifications and alerts. It will notify users about orders, payments, and delivery updates. Timely information delivery will be ensured through this module. It will improve communication between the system and users.

10. Security Mechanism

- **User Authentication**
  Secure login will be enforced using username and password validation. Only authenticated users will be allowed to access system features.

- **Role-Based Access Control**
  Access to system functionalities will be restricted based on user roles. Customers, vendors, and administrators will be granted different permissions.

- **Password Protection**
  User passwords will be stored in encrypted or hashed form. This will prevent unauthorized access to sensitive login information.

- **Secure Data Handling**
  User, order, and payment data will be handled securely within the system. Proper validation will be applied before storing or processing data.

- **Session Management**
  User sessions will be managed securely after successful login. Automatic session termination will help prevent misuse of inactive sessions.

- **Input Validation**
  User inputs will be validated to prevent invalid or malicious data entry. This will reduce risks such as data manipulation and system misuse.

11. Testing

➢ Black Box Testing

Black box testing will verify the functional behaviour of PetZio without considering internal code logic. Test cases will be designed to check features such as login, product search, and order placement. Inputs will be provided through the user interface and outputs will be observed. This testing will ensure correct system response for valid and invalid user actions. It will confirm that all functional requirements of the pet supplies platform are met.

➢ White Box Testing

White box testing will focus on internal logic used in PetZio modules.
Code paths related to authentication, order processing, and payment handling will be examined. Conditional statements and loops will be tested for correct execution. This testing will help identify logical flaws in backend processing. It will improve the reliability and efficiency of the application code.

➢ Unit Testing

Unit testing will involve testing individual modules of PetZio separately. Modules such as user management, product management, and order handling will be tested independently. Each unit will be verified for correct input processing and output generation. This testing will be performed during development to detect errors early. It will ensure stability of each component before system integration.

NAME : TANNU
COURSE : BCA

12.Gantt Chart

Week 1 to 3

PetZio			
Tannu	Project start : Thu, 1 Jan 2026		
2351641695	Display week : 5		
TASK	PROGRESS	START	END
IDE Setups	0%	01-Jan-26	03-Jan-26
Website Design	0%	04-Jan-26	17-Jan-26
Database Design	0%	18-Jan-26	21-Jan-26
Coding	0%	22-Jan-26	23-Jan-26
Testing	0%	24-Feb-26	27-Feb-26
Deployment	0%	28-Feb-26	03-Mar-26

01-Jan-26							08-Jan-26							15-Jan-26						
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W

Week 4 to 6

PetZio			
Tannu	Project start : Thu, 1 Jan 2026		
2351641695	Display week : 5		
TASK	PROGRESS	START	END
IDE Setups	0%	01-Jan-26	03-Jan-26
Website Design	0%	04-Jan-26	17-Jan-26
Database Design	0%	18-Jan-26	21-Jan-26
Coding	0%	22-Jan-26	23-Jan-26
Testing	0%	24-Feb-26	27-Feb-26
Deployment	0%	28-Feb-26	03-Mar-26

22-Jan-26							29-Jan-26							05-Feb-26						
22	23	24	25	26	27	28	29	30	31	1	2	3	4	5	6	7	8	9	10	11
T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W

Week 6 to 9

PetZio			
Tannu	Project start : Thu, 1 Jan 2026		
2351641695	Display week : 5		
TASK	PROGRESS	START	END
IDE Setups	0%	01-Jan-26	03-Jan-26
Website Design	0%	04-Jan-26	17-Jan-26
Database Design	0%	18-Jan-26	21-Jan-26
Coding	0%	22-Jan-26	23-Jan-26
Testing	0%	24-Feb-26	27-Feb-26
Deployment	0%	28-Feb-26	03-Mar-26

12-Feb-26							19-Feb-26							26-Mar-26						
12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	1	2	3	4
T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W

13.Limitations of Project

- The system will depend on stable internet connectivity for smooth operation. Performance may be affected in areas with poor network access.

- The platform will support limited payment options in the initial phase. Advanced payment methods may not be available at the beginning.

- Real-time product availability may not always reflect instant stock changes. Delays in inventory updates can occur under high user activity.

- The system will have limited support for third-party integrations initially. Features such as analytics and recommendation services may be minimal.

- The application will be designed for standard user loads in early stages. Extremely high traffic may impact performance without further optimization.

14.Future Scope of Project :-

- Advanced recommendation features can be added to suggest pet products based on user preferences. This will improve personalized shopping experience for customers.

- Multiple and secure digital payment options can be integrated in future versions. This will enhance convenience and flexibility in transaction processing.

- Real-time inventory management can be implemented for accurate stock updates. This will reduce product availability issues and improve vendor management.

- A mobile application version of the platform can be developed. This will increase accessibility and user engagement across devices.

- AI-based chat support can be integrated to assist users with queries and product guidance. This will improve customer support efficiency and response time.

- Advanced analytics and reporting features can be introduced for vendors and administrators. This will help in better decision-making and system monitoring.

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